F79SP Session 13: Limiting distributions

1 Limiting distribution

Definition 1.1. *Let $\boldsymbol{\mu} = (\mu_i)_{i\in S}$ be an initial distribution of a Markov chain $\{X_n\}_{n\geq 0}$ with transition matrix P. Suppose that there exists a distribution $\boldsymbol{\omega_\mu} = (\omega_{\mu,i})_{i\in S}$ such that:*

$$\boldsymbol{\omega_\mu} = \lim_{n\to\infty} \boldsymbol{\mu} P^n \qquad \textit{or equivalently} \qquad \omega_{\mu,j} = \lim_{n\to\infty} \sum_{i\in\mathcal{S}} \mu_i p_{i,j}^{(n)},\ j \in S. \tag{1}$$

*Then, $\boldsymbol{\omega_\mu}$ is called an associated **limiting distribution** of $\boldsymbol{\mu}$.*

In particular, fix any $i \in \mathcal{S}$. We consider the initial distribution $\boldsymbol{\mu}$ with 1 at state i, 0 otherwise. Then its limiting distribution (if exists) $\boldsymbol{\omega}_i$ becomes:

$$\omega_{ij} = \lim_{n\to\infty} \sum_{i\in\mathcal{S}} \mu_i p_{i,j}^{(n)} = \lim_{n\to\infty} p_{i,j}^{(n)}$$

The vector $\boldsymbol{\omega}_i = (\omega_{ij})_{j\in\mathcal{S}}$ is called a **limiting distribution** for initial state i.

2 Irreducibility and aperiodicity

We discuss two important properties of Markov chains that allows nice characterisation on their stationary and limiting distributions.

Irreducibility

A Markov chain with state space S being a single closed class is said to be **irreducible**.

Theorem 2.1. *Consider an irreducible THMC with stationary distribution $\pi = (\pi_j)_{j\in S}$. Then, for each j, $\pi_j > 0$.*

Proof. Outline as follows:

- Since $\pi P^n = \pi$, we show that $\pi_j = \sum_{i\in S} \pi_i p_{ij}^{(n)} > 0$.
- As $\sum_{j\in S} \pi_j = 1$ and each term in summation is non-negative, atleast one of them, says $\pi_k > 0$.
- As S is a single class, $k \to j$ for each j, $p_{kj}^{(n)} > 0$ for some n.
- Together we have $\pi_k p_{kj}^{(n)} > 0$, and this is a term in summation of $\pi_j = \sum_{i\in S} \pi_i p_{ij}^{(n)}$, resulting in $\pi_j > 0$.

□

Result: For irreducible THMC, if S is finite, stationary distribution does exists and is unique.

Aperiodicity (inside a closed class)

Definition 2.1. *An irreducible Markov chain* $\{X_n\}_{n\geq 0}$ *is* ***periodic with period*** $d \geq 2$ *if and only if*
(a) The state space $\mathcal{S}$ *may be represented as a union of* d *disjoint sets,*

$$\mathcal{S} = \mathcal{S}_1 \bigcup \mathcal{S}_2 \bigcup \ldots \bigcup \mathcal{S}_d$$

such that, for any $i = 1, 2, \ldots, d$ *and for any* $x \in \mathcal{S}_i$,

$$\mathbf{P}(X_1 \in \mathcal{S}_{i+1} \mid X_0 = x) = 1$$

(here $\mathcal{S}_{d+1} = \mathcal{S}_1$*), and*
(b) d *is the biggest number for which such a splitting exists.*

Definition 2.2. *An irreducible Markov chain* $\{X_n\}_{n\geq 0}$ *is* ***aperiodic*** *if it is not periodic with any* $d \geq 2$.

Example 2.1. *Determine the period of Markov chain with following transition matrix:*

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{2}{3} & 0 & 0 \\ \frac{4}{5} & 0 & \frac{1}{5} & 0 & 0 \end{pmatrix}$$

Food for thought: Are SRWs aperiodic?

How to determine whether an irreducible MC is aperiodic?

Here are two sufficient conditions for aperiodicity.

1. If there exists $j \in \mathcal{S}$ such that $p_{jj} > 0$, then the Markov Chain is aperiodic.
2. If there exists $j \in \mathcal{S}$ such that, for some $n \geq 1$, both $p_{jj}^{(n)} > 0$ and $p_{jj}^{(n+1)} > 0$, then the Markov Chain is aperiodic.

Theorem 2.2. *Consider an irreducible Markov chain $\{X_n\}_{n\geq 0}$ with transition matrix P. Then $\{X_n\}_{n\geq 0}$ is **aperiodic** if and only if there exists some $n > 0$ such that P^n has all positive entries, or equivalently, $p_{ij}^{(n)} > 0$ for all $i, j \in \mathcal{S}$.*

Proof. Outline as follows. One direction only ($\Leftarrow$):

- Assume there exists $n > 0$ such that $p_{ij}^{(n)} > 0$ for all $i, j \in \mathcal{S}$, we show that $p_{ij}^{(n+1)} > 0$, and hence MC must be aperiodic by Condition 2. above.
- Using Chapman-Kolmogorov equations:

$$p_{ij}^{(n+1)} = \sum_{k\in S} p_{ik}^{(1)} p_{kj}^{(n)} = \sum_{k\in S} p_{ik} p_{kj}^{(n)}$$

- As i must go to some state k_0 in one time-step ($\sum_k p_{ik}^{(m)} = 1$), there exists $k_0 \in S$ such that $p_{ik_0} > 0$.
- Hence, the corresponding term $p_{ik_0} p_{k_0 j}^{(n)}$ in:

$$p_{ij}^{(n+1)} = \sum_{k\in S} p_{ik} p_{kj}^{(n)}$$

must be positive, hence $p_{ij}^{(n+1)} > 0$.

□

Exercise: Show that MC in Example 1.2 (Session 12) is irreducible and aperiodic.

A useful theorem

Below theorem will be useful for Tutorial 4 attempts, which we will cover in more detail next week.

Theorem 2.3. *Suppose that a Markov chain is with below properties:*

1. *of finite state space S*
2. *S forms a single closed class (irreducible)*
3. *aperiodic*

Then there exists a unique stationary distribution and this is always the associated limiting distribution of any initial distribution.